Candidate's name._____**Sign** ____

Personal number _____

UACE CHEM P 2 **SET 11**

EDUCATOR'S FORUM

Uganda Advanced Certificate of Education

Uganda Advanced Certificate of Education

CHEMISTRY PRACTICAL (PAPER 2)

Time: 3 hours 15 minutes

INSTRUCTIONS TO CANDIDATES:

- This examination paper consists of two compulsory examination items.
- Responses to these items must be written in the answer booklet provided.
- Begin each item on a fresh page.
- All working must be clearly shown. Graph papers will be provided.

For examiners use only:

COMPETENCE	P	M	Dp	Da	O	C&R
ITEM 1						
ITEM 2						

ITEM 1

During a fuel-blending project at a research laboratory in Kampala, a volatile liquid byproduct code-named A2 was collected from the anaerobic fermentation of sweet potato waste. Initial tests indicate it is highly flammable and could serve as a clean alternative household fuel or industrial solvent. The lead chemist believes the molecule belongs to the aliphatic alcohol family but needs to establish its structural classification (primary, secondary, or tertiary) and saturation status to determine its combustion efficiency. You are provided with 10 cm^3 of liquid A2.

Task

Design and carry out a systematic qualitative analysis investigation to identify the functional group family, classification, and carbon backbone saturation level of sample A2. A minimum of 6 distinct diagnostic tests . Based on your findings, write its IUPAC name, structural formula, and explain its reactive properties in standard laboratory environments.

ITEM 2

An agricultural cooperative stores large amounts of ammonium sulphate fertiliser in a humid warehouse. Farmers recently complained that the fertiliser seemed less effective at boosting crop yields. The warehouse manager suspects that partial decomposition or moisture absorption has reduced the nitrogen content below the minimum regulatory standard of 20.5% Nitrogen by mass. A local school lab has been requested to verify the quality of the damaged fertiliser batch using a back-titration method where a sample is heated with excess sodium hydroxide, and the remaining alkali is determined.

Provided Materials: You are provided with the following materials:

- **FA1:** 0.10 mol dm^{-3} standard hydrochloric acid
- **FA2:** Sodium hydroxide solution (approx. 0.20 mol dm^{-3})
- **Solid Y2:** 1.50 g of the crushed ammonium sulphate fertiliser sample
- **Indicator:** Methyl red indicator
- **Equipment:** Access to a heating source / water bath
- Relevant reactions
- Reaction of Ammonium Sulphate (Solid Y_2) by Heating with Excess Sodium Hydroxide

$$(NH_4)_2SO_4(aq) + 2NaOH(aq) \rightarrow Na_2SO_4(aq) + 2NH_3(g) + 2H_2O(l)$$

- Back-Titration of Unreacted (Excess) Sodium Hydroxide with Hydrochloric Acid (FA1)

$$HCl(aq) + NaOH(aq) \rightarrow NaCl(aq) + H_2O(l)$$

Task

Design and carry out a quantitative back-titration experiment to determine the percentage of nitrogen by mass in Solid Y2. Present your titration data in a clear table showing at least six accurate titre values, and show all steps of your calculations, including equations for the reactions involved. Formulate an appropriate recommendation for the cooperative regarding the quality of the fertiliser batch.

END